

Assignment 1

Daria-Mihaela Broscotanu - AI 407

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(1, 2) /
 (3, 4)
 (2, 5, 3, 5)
 $a = 2, 5$
 (3, 5, 4, 5)
 $a = 3, 5$

1

- a finite hypothesis class $\mathcal{H}$ with $\text{VCdim}(\mathcal{H}) = 2024$. Justify your choice. (0.25 points)
- an infinite hypothesis class $\mathcal{H}$ with $\text{VCdim}(\mathcal{H}) = 2024$. Justify your choice. (0.25 points)

1.1

As presented in Seminar 4, exercise 1, a finite hypothesis class is the one of monotone Boolean conjunctions over $\{0, 1\}^d$. Setting the value of d to be 2024, in order to match the requirement of having the $\text{VCdim}(\mathcal{H}_{\text{mon}}^{2024}) = 2024$, results in the following finite hypothesis class structure:

$$\mathcal{H}_{\text{mon}}^{2024} = \{0, 1, x_1, x_2, \dots, x_{2024}, x_1 \wedge x_2, \dots, x_1 \wedge x_2 \wedge \dots \wedge x_{2024}\}$$

From the seminar, we already know that $\mathcal{H}_{\text{mon}}^{2024}$ is a finite class, with it having the $|\mathcal{H}_{\text{mon}}^{2024}| = 2^{2024} + 1$.

This is proven by taking the set $C = \{c_1, \dots, c_{2024}\}$, with $c_i = (1, 1, \dots, 1) - e_i$, $i = \overline{1, 2024}$ and proving that for each of the possible labeling of these points, we can find the corresponding h . Basing this on the proof provided in the seminar 4, we can find the set C that shatters $\mathcal{H}_{\text{mon}}^{2024}$.

1.2

If we look at homogenous linear classifiers, the ones with $b = 0$, presented in lecture 7, slide 20, we know from the proof presented during the course that the $\text{VCdim}(HS_0^n) = n$. Replacing the value of n with 2024, we achieve the required $\text{VCdim} = 2024$.

As each linear classifier can be defined as the product of a weight and a bias over R^{2024} , a h in our selected hypothesis class H looks like this: $h(x) = \text{sign}(\text{weight} * x + \text{bias})$, $\forall x \in R^{2024}$. The class in question is infinite by its definition over R^{2024} , including every possible combination of weight and bias pairs, resulting in infinitely many possible linear classifiers.

2

The area inside the triangle ABC can define the class of concepts that contains the classifiers that assign a sample to the positive class if said sample is placed in the triangle defined by the two catheti AB and AC and the line hypotenuse BC.

$$H_{ABC}^2 = \left\{ \begin{array}{l} h_{ABC}: R^2 \rightarrow \{0, 1\}, A = (a_1, a_2), B = (b_1, a_2), C = (a_1, c_2) \\ h_{ABC}(x_1, x_2) = \begin{cases} 1, & \text{if } x_1 \in [a_1, b_1] \text{ and } x_2 \in [a_2, c_2], \text{ and} \\ & \frac{(x_1 - a_1) \cdot (a_2 - c_2)}{(x_2 - c_2) \cdot (b_1 - a_1)} < 1 \\ 0, & \text{otherwise} \end{cases} \end{array} \right\}$$

First, we need to find the algorithm A .

We are under the realizability assumption, so there exists a labeling function $f \in H$, $f = h_{(a_1^*, b_1^*, a_2^*, b_2^*)}$ that labels the training data.

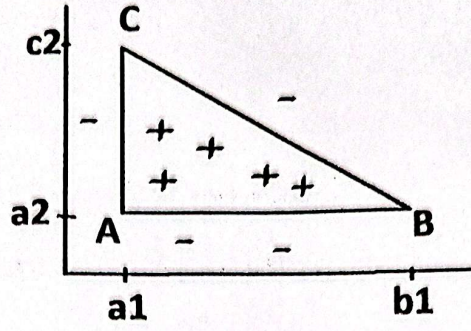


Figure 1: All the points that fall in triangle R^* will be labeled by h^* with label 1 (+), the other points will be labeled with label 0 (-).

Consider the training set $S = \left\{ (x_1, y_1), (x_2, y_2), \dots, (x_m, y_m) \mid \begin{array}{l} y_i = h^*_{(a_1^*, a_2^*, b_1^*, a_2^*, a_1^*, c_2^*)}(x_i) \\ x_i \in R, x_i = (x_{i1}, x_{i2}) \end{array} \right\}$

As in Figure 1, h^* labels each point drawn from the triangle defined by the points $A(a_1^*, a_2^*)$, $B(b_1^*, a_2^*)$, $C(a_1^*, c_2^*)$ with label 1, and all other points with label 0. So we have $h^*_{(a_1^*, a_2^*, b_1^*, a_2^*, a_1^*, c_2^*)} = 1_{R^*}$

Consider the following algorithm A , that takes as input the training set S and outputs h_S .

$h_S = h_{(a_{1S}, a_{2S}, b_{1S}, a_{2S}, a_{1S}, c_{2S})}$, where

$$\begin{aligned} a_{1S} &= \min_{\substack{i=1, \dots, m \\ y_i=1}} x_{i1} & a_{2S} &= \min_{\substack{i=1, \dots, m \\ y_i=1}} x_{i2} \\ b_{1S} &= \max_{\substack{i=1, \dots, m \\ y_i=1}} x_{i1} & c_{2S} &= \max_{\substack{i=1, \dots, m \\ y_i=1}} x_{i2} \end{aligned}$$

If all $y_i = 0$, then all points x_i have label 0, so there is no positive example. In this case, choose $z = (z_1, z_2)$ a point that is not in the training set S and take $a_{1S} = b_{1S} = z_1$, $a_{2S} = c_{2S} = z_2$.

As it is defined to contain all the positive labeled samples, by construction, A is an ERM, meaning that $L_{h^*, H}(h_S) = 0$, h_S doesn't make any errors on the training set S .

Now we want to find the sample complexity $m_H(\epsilon, \delta)$ such that

$$\mathbb{P}_{S \sim \mathcal{D}^m} (L_{h^*, \mathcal{D}}(h_S) \leq \epsilon) \geq 1 - \delta \text{ where } S \text{ contains } m \geq m_H(\epsilon, \delta) \text{ examples.}$$

We make the observation that h_S makes errors in region $R^* \setminus R_S$, assigning the label 0 to points that should get label 1. All points $\in R_S$ will be labeled correctly (label 1), all points outside R^* will be labeled correctly (label 0).

Let's fix $\epsilon > 0, \delta > 0$ and consider a distribution $\mathcal{D}$ over R^2 .

Case 1. If $\mathcal{D}(R^*) \leq \epsilon \Rightarrow L_{h^*, \mathcal{D}}(h_S) = \mathbb{P}_{x \sim \mathcal{D}} (h_S(x) \neq h^*(x)) = \mathbb{P}_{x \sim \mathcal{D}} (x \in R^* \setminus R_S) \leq \mathbb{P}_{x \sim \mathcal{D}} (x \in R^*) \leq \epsilon \Rightarrow P = 1$

Case 2. If $\mathcal{D}(R^*) > \epsilon \Rightarrow$

We have two subcases: 1. In case of overlapping:

$$\begin{aligned} L_{h^*, \mathcal{D}}(h_S) &= \mathbb{P}_{x \sim \mathcal{D}} (h^*(x) \neq h_S(x)) = \mathbb{P}_{x \sim \mathcal{D}} (x \in R^* \setminus R_S) \leq \mathbb{P}_{x \sim \mathcal{D}} (x \in R_1 \cup R_2 \cup R_3) \leq \\ &\leq \sum_{i=1}^3 \mathbb{P}_{x \sim \mathcal{D}} (x \in R_i) = \sum_{i=1}^3 \mathcal{D}(R_i) = 3 \cdot \frac{\epsilon}{3} = \epsilon \Rightarrow P = 1 \end{aligned}$$

2. In case of no overlapping:

$$P(L_{h^*, \mathcal{D}}(h_S) > \epsilon) \leq 3 * (1 - \frac{\epsilon}{3})^m < \delta \Rightarrow m > \frac{3}{\epsilon} \log \frac{3}{\delta}$$

This implies H is PAC-learnable.

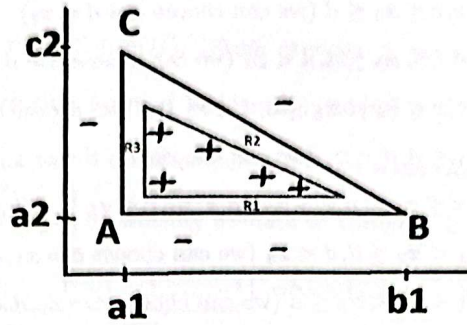


Figure 2: R1, R2, R3

3

$$\mathcal{H}_1 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = 1_{[x \geq a]}(x) = 1_{[a, +\infty)}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_2 = \{h_b : \mathbb{R} \rightarrow \{0, 1\} \mid h_b(x) = 1_{[x < b]}(x) = 1_{(-\infty, b)}(x), b \in \mathbb{R}\},$$

$$\mathcal{H}_3 = \{h_{c,d} : \mathbb{R} \rightarrow \{0, 1\} \mid h_{c,d}(x) = 1_{[c \leq x \leq d]}(x) = 1_{[c,d]}(x), c, d \in \mathbb{R}\}.$$

$$\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{H}_3$$

Let's compute the $VCdim$ for each hypothesis class $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$.

The intuition (from seminar 3, exercise 4) for $VCdim(\mathcal{H}_1) = 1$, we will prove that for two points $x_1, x_2 \in \mathbb{R}, x_1 < x_2$, the all possible labelings are not achievable:

- (0, 0): we choose $x_2 < a$
- (0, 1): we choose $x_1 < a \leq x_2$
- (1, 0): we have to pick $a < x_1$, but $a > x_2$ (impossible)
- (1, 1): we choose $a \leq x_1$

So $VCdim(\mathcal{H}_1) = 1$.

For $\mathcal{H}_2$, we will prove that for two points $x_1, x_2 \in \mathbb{R}, x_1 < x_2$, the all possible labelings are not achievable:

- (0, 0): we choose $b < x_1$
- (0, 1): we have to pick $b \leq x_1$, but $b > x_2$ (impossible)
- (1, 0): we choose $x_1 < b \leq x_2$
- (1, 1): we choose $x_2 < b$

So $VCdim(\mathcal{H}_2) = 1$.

For $\mathcal{H}_3$, we will prove that for two points $x_1, x_2 \in \mathbb{R}, x_1 < x_2$, the all possible labelings are achievable:

- (0, 0): we choose $x_2 < c < d < x_1$
- (0, 1): we choose $c < x_1, c \leq x_2 \leq d$ (we can choose $c = d = x_2$)
- (1, 0): we choose $c \leq x_1 \leq d, d < x_2$ (we can choose $c = d = x_1$)
- (1, 1): we choose $c \leq x_1 < x_2 \leq d$

So $VCdim(H_3) \geq 2$.

Next we will prove that for three points $x_1, x_2, x_3 \in R, x_1 < x_2 < x_3$, the all possible labelings are NOT achievable:

- (0, 0, 0): we choose $x_3 < c < x_1$
- (0, 0, 1): we choose $x_2 < c, c \leq x_3 \leq d$ (we can choose $c = d = x_3$)
- (0, 1, 0): we choose $x_1 < c, c \leq x_2 \leq d, d < x_3$ (we can choose $c = d = x_2$)
- (0, 1, 1): we choose $x_1 < c, c \leq x_2 < x_3 \leq d, d < x_3$ (we can choose $c = x_2, d = x_3$)
- (1, 0, 0): we choose $c \leq x_1 \leq d, d < x_2$ (we can choose $c = d = x_1$)
- (1, 0, 1): we have to pick $c \leq x_1 \leq d, c > x_2$ or $d < x_2, c \leq x_3 \leq d$ (impossible)
- (1, 1, 0): we choose $c \leq x_1 < x_2 \leq d, d < x_3$ (we can choose $c = x_1, d = x_2$)
- (1, 1, 1): we choose $c \leq x_1 < x_2 < x_3 \leq d$ (we can choose $c = x_1, d = x_3$)

So $VCdim(H_3) = 2$.

For the combined hypothesis class H , it can be easily seen that $VCdim(H) \leq 1$, due to the information above.

For two points $x_1, x_2 \in R, x_1 < x_2$:

- (0, 0): we choose $H_1 : x_2 < a$ or $H_2 : b < x_1$ or $H_3 : x_2 < c$ or $x_1 > d$
- (0, 1): we choose $H_1 : x_2 \geq a$ or $H_3 : x_1 < c \leq x_2 \leq d$
- (1, 0): we choose $H_2 : x_1 < b$ or $H_3 : c \leq x_1 \leq d < x_2$
- (1, 1): we choose $H_1 : x_1 \geq a$ or $H_2 : b > x_2$ or $H_3 : c \leq x_1 < x_2 \leq d$

So $VCdim(H) \geq 2$.

For three points $x_1, x_2, x_3 \in R, x_1 < x_2 < x_3$:

- (0, 0, 0): we choose $H_1 : x_3 < a$ or $H_2 : b < x_1$ or $H_3 : x_3 < c$ or $x_1 > d$
- (0, 0, 1): we choose $H_1 : x_3 \geq a$ or $H_3 : x_2 < c \leq x_3 \leq d$
- (0, 1, 0): we choose $H_3 : x_1 < c \leq x_2 \leq d < x_3$
- (0, 1, 1): we choose $H_1 : x_2 \geq a$ or $H_3 : x_1 < c \leq x_2 < x_3 \leq d$
- (1, 0, 0): we choose $H_2 : x_1 < b$ or $H_3 : c \leq x_1 \leq d < x_2 < x_3$
- (1, 0, 1): impossible
- (1, 1, 0): we choose $H_2 : x_2 < b$ or $H_3 : c \leq x_1 < x_2 \leq d < x_3$
- (1, 1, 1): we choose $H_1 : x_1 \geq a$ or $H_2 : x_3 < b$ or $H_3 : c \leq x_1 < x_2 < x_3 \leq d$

So $VCdim(H) = 2$.

4

(1 point) Consider $\mathcal{H}$ the class of 3-piece classifiers (signed intervals): $\mathcal{H} = \{h_{a,b,s} : R \rightarrow \{-1, 1\} \mid a \leq b, s \in \{-1, 1\}\}$, where
$$h_{a,b,s}(x) = \begin{cases} s, & x \in [a, b] \\ -s, & x \notin [a, b] \end{cases}$$

4.1 a

First, we have to compute the $VCdim(H)$. From seminar 3, exercise 4, we already proved that $VCdim(H) = 3$.

From the lecture, we know that for $m \leq VCdim(H)$, $\tau_H(m) = 2^m$, so next we need to find what happens for $m \geq 4$.

With $\tau_H(m)$ being the number of different function from a set C_m to $\{0, 1\}$, we consider the set $C_m = \{x_1, x_2, \dots, x_m\}$, with $x_1 < x_2 < \dots < x_m$.

We have to compute four types of labeling formats to compute in order to find out the value of $\tau_H(m)$, for $m \geq 4$:

- A: $(\underbrace{-1, -1, \dots, -1}_{k_1}, \underbrace{1, 1, \dots, 1}_{k_2}, \underbrace{-1, -1, \dots, -1}_{k_3})$, where $k_1 + k_2 + k_3 = m$.
- B: $(\underbrace{1, 1, \dots, 1}_{k_1}, \underbrace{-1, -1, \dots, -1}_{k_2}, \underbrace{1, 1, \dots, 1}_{k_3})$, where $k_1 + k_2 + k_3 = m$.
- C: $(\underbrace{-1, -1, \dots, -1}_{k_1}, \underbrace{1, 1, \dots, 1}_{k_2})$, where $k_1 + k_2 = m, k_2 > 0$.
- D: $(\underbrace{1, 1, \dots, 1}_{k_1}, \underbrace{-1, -1, \dots, -1}_{k_2})$, where $k_1 + k_2 = m, k_2 > 0$.

To assess how many labelings are possible, we will take some examples for case A:

- $k_2 = 0 \Rightarrow 1$ labeling - $(-1, -1, \dots, -1)$
- $k_2 = 1 \Rightarrow m$ labelings - we can place the 1 on each of the m positions
- $k_2 = 2 \Rightarrow m - 1$ labelings - we can place the two consecutive ones on each pair of positions $(i, i + 1), \forall i \in 1, m - 1$
- ...
- $k_2 = m \Rightarrow 1$ labeling - we have only ones in the labeling

So, for case A, we have $A = 1 + \frac{m \cdot (m+1)}{2}$.

To assess how many labelings are possible, we will take some examples for case B:

- $k_2 = 0 \Rightarrow 1$ labeling - $(1, 1, \dots, 1)$
- $k_2 = 1 \Rightarrow m$ labelings - we can place the -1 on each of the m positions
- $k_2 = 2 \Rightarrow m - 1$ labelings - we can place the two consecutive -1s on each pair of positions $(i, i + 1), \forall i \in 1, m - 1$
- ...
- $k_2 = m \Rightarrow 1$ labeling - we have only -1 in the labeling

So, for case B, we have $B = 1 + \frac{m \cdot (m+1)}{2}$.

To assess how many labelings are possible, we will take some examples for case C:

- $k_2 = 1 \Rightarrow 1$ labeling - $(-1, -1, \dots, 1)$
- $k_2 = 2 \Rightarrow 1$ labeling - $(-1, -1, \dots, 1, 1)$
- ...
- $k_2 = m \Rightarrow 1$ labeling - $(1, 1, \dots, 1)$

So, for case C, we have $C = m$. The same conclusion can be drawn for $D = m$.

As for both C and D, $k_2 > 0$, we can easily see that our sets of labeling do not have common points, as they have a reverse structure of the 1s and -1s.

For C and D, the result of their reunion $E = C \cup D$ would imply only the concatenation of the two sets, so we can be sure that $|E| = m + m = 2m$.

Based on the results above, the result for growth function is computed as follows: $\tau_H(m) = |A| + |B| - |E| = 2 * (1 + \frac{m \cdot (m+1)}{2}) - 2m = m * (m + 1) - 2m + 2 = m * (m - 1) + 2$.

4.2 b

From section 4.1, the growth function is:

$$\tau_H(m) = \begin{cases} 2^m, & \text{if } m \leq 3, \\ m * (m - 1) + 2, & \text{if } m > 3. \end{cases}$$

From lecture 7, we know from Sauer - Shelah - Perles lemma that for H with a $VCdim(H) = d < \infty$, we have:

$$\tau_H(m) \leq \sum_{i=0}^d \binom{m}{i}$$

In our case $d = 3$, so we have the following:

$$\sum_{i=0}^3 \binom{m}{i} = \binom{m}{0} + \binom{m}{1} + \binom{m}{2} + \binom{m}{3} = 1 + m + \frac{m \cdot (m-1)}{2} + \frac{m \cdot (m-1) \cdot (m-2)}{6}$$

$$m(m-1) + 2 \leq 1 + m + \frac{m \cdot (m-1)}{2} + \frac{m \cdot (m-1) \cdot (m-2)}{6} \cdot 6$$

$$6m(m-1) + 12 \leq 6 + 6m + 3m(m-1) + m(m-1)(m-2)$$

$$0 \leq 6 + 6m + 3m(m-1) + m(m-1)(m-2) - 6m(m-1) - 12$$

$$0 \leq (m-1)(m^2 - 5m + 6)$$

$$0 \leq (m-1)(m-2)(m-3), \forall m > 3$$

- $m = 0 : 0 \leq -6$
- $m = 1, m = 3, m = 3 : 0 \leq 0$
- $m > 0$: the inequality is strict

4.3 c

Yes, such class exists. $H_{thresholds}$ is an example of a hypothesis class for which the shattering coefficient $\tau_{H_{thresholds}}(m)$ is equal to the general upper bound. This is because we know from lecture 7 that $\tau_{H_{thresholds}}(m) = m + 1 = \sum_{i=0}^1 \binom{m}{i} = \binom{m}{0} + \binom{m}{1} = 1 + m$.

5

$$\mathcal{H} = \{h : \mathbb{R} \rightarrow \{0, 1\} \mid h_\theta(x) = 1_{[\theta, \theta+1] \cup [\theta+2, \infty)}(x), \theta \in \mathbb{R}\}$$

To compute the $VCdim(H)$ we will use the following cases in order to see the number of points we can shatter using this class.

1. Taking $\forall x \in \mathbb{R}$, for any given number of θ , we can compute the label $h(x)$, as it would take the value 1 for $x \in [\theta, \theta + 1] \cup [\theta + 2, \infty)$, or 0 otherwise. So our H can be shattered by one point, regardless of θ .

2. To see if our H can be shattered by 2 points, we should take two points, $a, b \in \mathbb{R}$, with $a < b$, and see if these two points can be classified using H .

For each of the following labelings:

- $(0, 0)$: θ can be chosen to place a and b in $(-\infty, \theta)$, so $\theta > b$
- $(0, 1)$: θ can be chosen to place a in $(-\infty, \theta)$ and b in $[\theta, \theta + 1] \cup [\theta + 2, \infty)$ so $a < \theta$ and $\theta \leq b \leq \theta + 1$
- $(1, 0)$: θ can be chosen to place a in $[\theta, \theta + 1]$ and b in $(\theta + 1, \theta + 2)$, so $\theta \leq a \leq \theta + 1 < b < \theta + 2$
- $(1, 1)$: θ can be chosen to place a in $[\theta, \theta + 1]$ and b in $[\theta + 2, \infty)$, so $\theta \leq a \leq \theta + 1 < \theta + 2 \leq b$

Example: $a = 1.5, b = 2.5$:

- $(0, 0)$: $\theta = 0$
- $(0, 1)$: $\theta = 2$

- (1, 0): $\theta = 1$
- (1, 1): $\theta = -1$

All possible labelings are covered, so $VCdim(H) \geq 2$.

3. To see if out H can be shattered by 3 points, we should take three points, $a, b, c \in R$, with $a < b < c$, and see if these to points can be classified using H .

- (0, 0, 0): θ can be chosen to place a, b, c in $(-\infty, \theta)$, so $\theta > c$ ($a = \theta - 3, b = \theta - 2, c = \theta - 1$)
- (0, 0, 1): θ can be chosen to place a, b in $(-\infty, \theta)$ and c in $[\theta, \theta + 1] \cup [\theta + 2, \infty)$ so $a < b < \theta$ and $\theta \leq c \leq \theta + 1$ (for example: $a = \theta - 2, b = \theta - 1, c = \theta$)
- (0, 1, 0): θ can be chosen to place a in $(-\infty, \theta)$, b in $[\theta, \theta + 1]$, c in $(\theta + 1, \theta + 2)$, (for example: $a = \theta - 1, b = \theta, c = \theta + 1.5$)
- (0, 1, 1): θ can be chosen to place a in $(-\infty, \theta)$, b in $[\theta, \theta + 1]$, c in $[\theta + 2, \infty)$, (for example: $a = \theta - 1, b = \theta, c = \theta + 2$)
- (1, 0, 0): θ can be chosen to place a in $[\theta, \theta + 1]$, b, c in $(\theta + 1, \theta + 2)$, (for example: $a = \theta, b = \theta + 1.5, c = \theta + 1.75$)
- (1, 0, 1): θ can be chosen to fit $\theta \leq a, \theta + 1 < b < \theta + 2, \theta + 2 \leq c$, (for example: $a = \theta, b = \theta + 1.5, c = \theta + 2$)
- (1, 1, 0): θ can be chosen to fit $\theta \leq a < b \leq \theta + 1, \theta + 1 < c < \theta + 2$, (for example: $a = \theta + 0.5, b = \theta + 0.75, c = \theta + 1.5$)
- (1, 1, 1): θ can be chosen to fit $\theta + 2 \leq a$, (for example: $a = \theta + 2, b = \theta + 3, c = \theta + 4$)

All possible labelings are covered, so $VCdim(H) \geq 3$.

4. To see if out H can be shattered by 4 points, we should take three points, $a, b, c \in R$, with $a < b < c < d$, and see if these to points can be classified using H . To see if out H can be shattered by 4 points, we should take four points, $a, b, c, d \in R$, with $a < b < c < d$, and see if these to points can be classified using H .

- (1, 0, 1, 0): θ should fit $\theta + 1 \leq a, \theta + 1 < b < \theta + 2, \theta + 2 \leq c$, (for $a = \theta, \theta + 1 < b < \theta + 2, \theta + 2 < c$, but $c < d$, and there are no more intervals after $(\theta + 2, \infty)$ where the label is 0, so this labeling is not achievable)

That means that $VCdim(H) \geq 3$.

Example: $a = 1, b = 2, c = 2.3$:

- (0, 0, 0): $\theta = 4$
- (0, 0, 1): $\theta = 2.3$
- (0, 1, 0): $\theta = 1.1$
- (1, 0, 0): $\theta = 0.5$
- (1, 1, 0): $\theta = 1$
- (1, 0, 1): $\theta = 0.3$
- (0, 1, 1): $\theta = 2$
- (1, 1, 1): $\theta = -1$

For 4 points, we cannot achieve (1,0,1,0), because we should have the following (with $a < b < c < d$):

- $a \in [\theta, \theta + 1]$
- $b \in (\theta + 1, \theta + 2)$
- $c \in [\theta + 2, \infty)$
- $c < d$, but d should have the label 0.

So $VCdim(H) = 3$.

From the hint, we know that if we want to show that the $VCdim$ of 1-decision lists over $\{0,1\}^n$, we have to represent these lists as halfspaces. We know that the value $L(a)$ is defined to be b_j where j is the smallest index satisfying $c_j(a) = 1$; if no such index exists, then $L(a) = b$. We want to convert this logic into a linear function we can use for classification.

We have $L = \{(x_1 \wedge \overline{x_3}, 0), (x_4, 1), (\overline{x_2} \wedge x_3, 1), (x_1 \wedge x_5, 0), (\overline{x_4} \wedge x_6, 1), (x_1 \wedge \overline{x_8}, 0)\}$.

From the lecture, we know that for H being the set of threshold half-spaces over $\mathbb{R}^n$, we have the $VCdim(H) = n + 1$, so we can pick $\alpha = \beta = \gamma = \delta = 1$.

References

(4 5 6)

(1, 2, 3)

$[-2, -1] \cup [0, 1]$

$$j = -9 \quad [-9, -8] \cup [-7, -5] \quad \checkmark$$

$$j = 6 \quad [6, 7] \cup [8, 10] \quad \checkmark$$

$$j = 4, 5 \quad [4, 5, 5, 5] \cup [6, 5, 8, 5] \quad \checkmark$$

$$j = 2, 5 \quad [2, 5, 3, 5] \cup [4, 5, 6, 5] \quad \checkmark$$

$$j = 0 \quad [0, 1] \cup [2, 4] \quad \checkmark$$

$$j = 3, 5 \quad [3, 5, 4, 5] \cup [5, 5, 7, 5] \quad \checkmark$$

$$j = 1 \quad [1, 2] \cup [3, 5] \quad \checkmark$$

$$j = 2 \quad [2, 3] \cup [4, 6] \quad \checkmark$$